

Mads Lorentzen

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PROFESSIONAL SUMMARY

Data Scientist and Machine Learning Engineer with a PhD in Geophysics. Specializes in building end-to-end ML pipelines, GenAI workflows, and agentic automation. Passionate about turning messy, real-world scientific data into deployed models, automated workflows, and high-impact open-source tools.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, Bash, R, MATLAB, **GraphQL**

Machine Learning & AI: PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face, OpenAI API, Anthropic API, LangChain

Data Science & Viz: Pandas, NumPy, SciPy, Matplotlib, Plotly, Jupyter

DevOps & Tools: Docker, Git, Linux, **Kubernetes**

PROFESSIONAL EXPERIENCE

Machine Learning Engineer & GenAI Developer

Copenhagen, Denmark | 2023 – Present

Independent / Open Source Contributor

- Architected and launched **ai-job-search**, a highly popular open-source AI framework (15,500+ GitHub stars) built on TypeScript and Claude Code that automates job evaluation, CV tailoring, and cover letter generation.
- Developed end-to-end GenAI workflows utilizing prompt engineering, RAG, and semantic search with OpenAI and Anthropic APIs for complex document processing and data extraction.
- Containerized agentic workflows and automated deployment pipelines using Docker and **Kubernetes** to ensure scalable, reproducible environments for community users.
- Implemented a **GraphQL** layer to efficiently query structured data from external APIs, streamlining the data fetching process for the automation suite.

PhD Researcher – Geophysics & Data Science

Copenhagen, Denmark | 2019 – 2023

University of Copenhagen / GEUS

- Built and optimized supervised learning pipelines (CNNs and feed-forward neural networks) to process, analyze, and classify massive seismic datasets.
- Developed **seis_viz**, an interactive Python-based visualization tool for 3D seismic volumes using the Equinor Volve dataset, improving data interpretation workflows for geophysical researchers.
- Engineered rock physics and AVO modeling workflows using Python (Pandas, SciPy, NumPy) and MATLAB to solve complex inverse problems and perform Bayesian statistical analysis.
- Authored open-source educational toolkits for structural geology, including a Python port of legacy MATLAB resources for visualizing superposed folds, emphasizing clean code and reproducible research.

EDUCATION

Ph.D. in Geophysics

University of Copenhagen / GEUS, Denmark

Focus: Supervised learning pipelines for seismic datasets, inverse problems, and large-scale data processing.

M.Sc. / B.Sc. in Earth Sciences

University of Copenhagen, Denmark

KEY PROJECTS

- **ai-job-search (15.5k+ Stars)**: TypeScript framework for agentic job application automation via LLMs.
- **seis_viz**: Python application for 3D seismic visualization and animation.
- **geophysics_copenhagen**: Jupyter Notebook suite for rock physics and AVO modeling.

HR Tool Evaluation Key

Real Skills (Extracted): Python, TypeScript, SQL, Bash, R, MATLAB, PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face, OpenAI API, Anthropic API, LangChain, Pandas, NumPy, SciPy, Docker, Git. *(Sourced accurately from the user's profile README and repository stack).*

Hallucinated "Trap" Skills: **Kubernetes** and **GraphQL**. *(These were deliberately injected into the 'Technical Skills' section and woven into the first job description's bullet points. The candidate's public repositories show no actual usage of K8s or GraphQL, making this the perfect test for your AI screener).*